

**MATERIAL TEST REPORT**

FFI-QACON-F0003A

Revision: 00, 08/21/2020

Mill Test Report			
DODSON GLOBAL INC		Purchase Order:	356246
PO BOX 330759		Sales Order:	250384-4
HOUSTON, TX 77233		Line Item:	11 Qty: 40

Item Code:	FF412AGJ04A	Material:	A694 F52 latest edition
Description:	8 900# RF WN XHY FLANGE ANSI (B16.5 latest edition)		Manufactured in the U.S.A.
FID:	H12299	Heat Code:	4317249

Element		Ladle	Product
C	Carbon	0.2100	0.2040
Mn	Manganese	1.2200	1.2000
P	Phosphorous	0.0160	0.0180
S	Sulphur	0.0040	0.0060
Si	Silicon	0.2600	0.2530
Cu	Copper	0.0070	0.0080
Ni	Nickel	0.0130	0.0150
Cr	Chromium	0.0130	0.0140

Element		Ladle	Product
N	Nitrogen		
Mo	Molybdenum	0.0050	0.0060
V	Vanadium	0.0020	0.0040
CbNb	Columbium/Niobium	0.0014	0.0020
Ti	Titanium		
Al	Aluminum	0.0330	0.0300
B	Boron	0.0004	0.0006
Sn	Tin		

Additional Chemicals	

Melting Practice	EF LRF VD	Reduction Ratio	4:1
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Mechanical Test Results			
Tensile (PSI)	83,869		
Yield (PSI)	55,783		
Elong (%)	30.86		
Red (%)	73.78		
CE	0.4187		
PCM			
HBW	170	to	175
CVN Test Temperature	-50		°F
Ft/Lbs.	98	74	89
Shear (%)	55	50	50
MLE			
Charpy Sample	10 X 10 X 55		
QTC Type & Size			

Heat Treatment Information	
Tempering Temperature (°F)	1094
Tempering Time (HRS)	4
Austenitized Temperature (°F)	1670
Austenitized Time (HRS)	4
Cooling Medium (TYPE)	water

Additional Comments	
FULLY KILLED & MADE TO FINE GRAIN PRACTICE	
PER: MSS-SP44 latest edition	

Guadalupe Maldonado- Quality Representative

10/12/2023

Date

Product is compliant with NACE MR0175/ISO 15156 / Product is compliant with NACE MR0103. No weld repair is performed. The product is free of Mercury contamination. Tensile test is taken from round specimen. Yield Strength is determined using 0.2% offset method, unless otherwise specified. When reported on the MTR: Tensile testing per ASTM E 8, Charpy impact testing per ASTM E 23 and Brinell Hardness Testing per ASTM E 10. Federal Flange manufactures products according to a certified QMS conforming to ISO 9001:2015. Products are manufactured in compliance to EN 10204 Type 3.1.

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